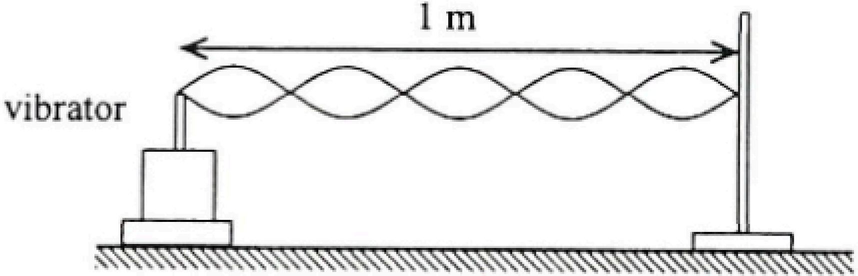
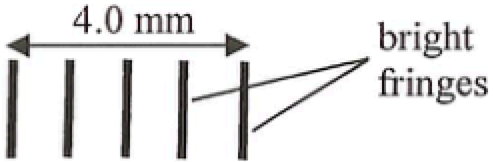
18. The figure shows a string with one end fixed and the other end tied to a vibrator. A stationary wave is formed as shown at a certain frequency.



If the speed of the wave along the string is 7 m s^{-1}, what is the frequency of the wave ?

A. 2.8 Hz
B. 7 Hz
C. 17.5 Hz
D. 35 Hz

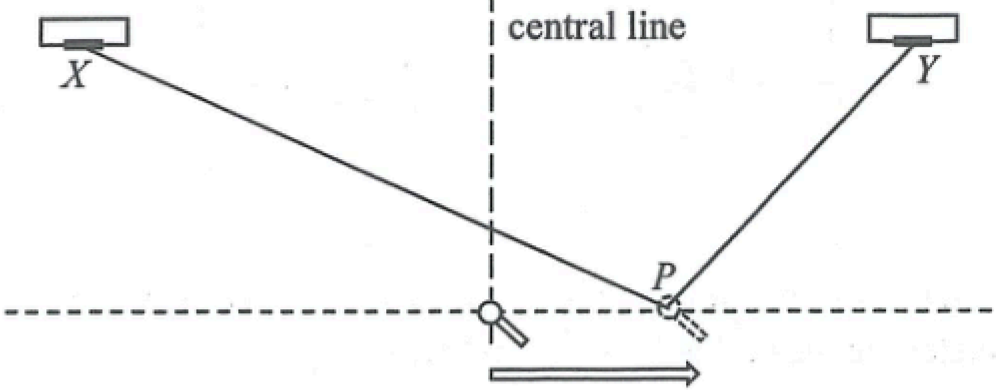
*17.



In Young's double-slit experiment, monochromatic light of wavelength λ is used and the slit separation is a. On a screen placed at a distance of 1.8 m from the double slits, the separation between five consecutive bright fringes is 4.0 mm. Which relationship below is correct ?

A. $a = 1800\ \lambda$
B. $a = 2250\ \lambda$
C. $a = 7200\ \lambda$
D. $a = 11250\ \lambda$

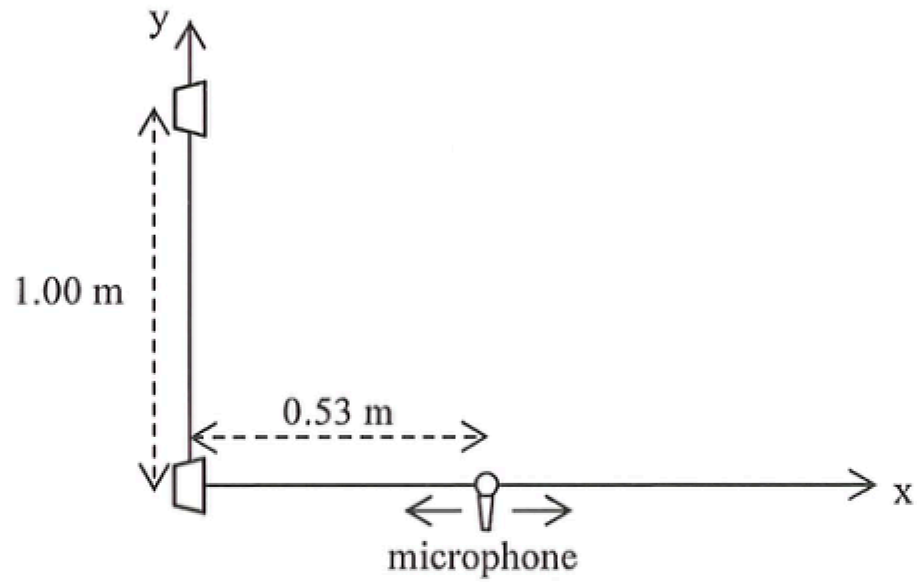
21.



Two loudspeakers X and Y emit sound waves of frequency 500 Hz. A microphone is moved steadily along a line perpendicular to the central line as shown. It detects sound waves of maximum amplitude at the central line and the next maximum amplitude is detected at point P. Find $PX - PY$.
Given: speed of sound in air = 340 m s^{-1}

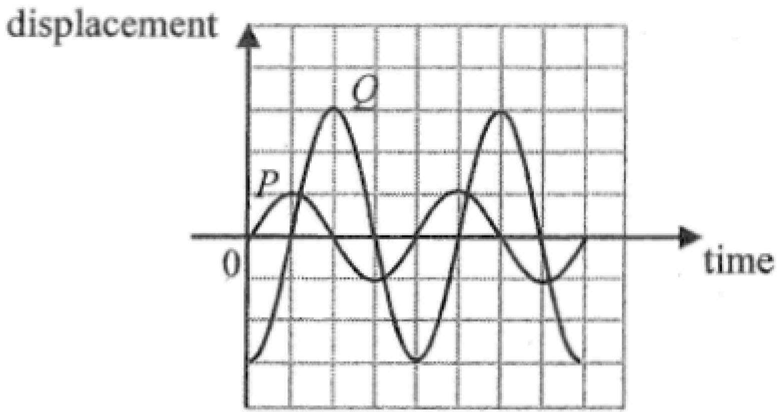
A. 0.17 m
B. 0.34 m
C. 0.51 m
D. 0.68 m

19. Two loudspeakers separated by 1.00 m along the y-axis emit coherent sound waves which are in phase. When a microphone is moved along the x-axis, only one maximum is detected at a distance of 0.53 m as shown. Find the wavelength of the sound waves.



A. 0.60 m
B. 0.75 m
C. 1.00 m
D. 1.20 m

13.

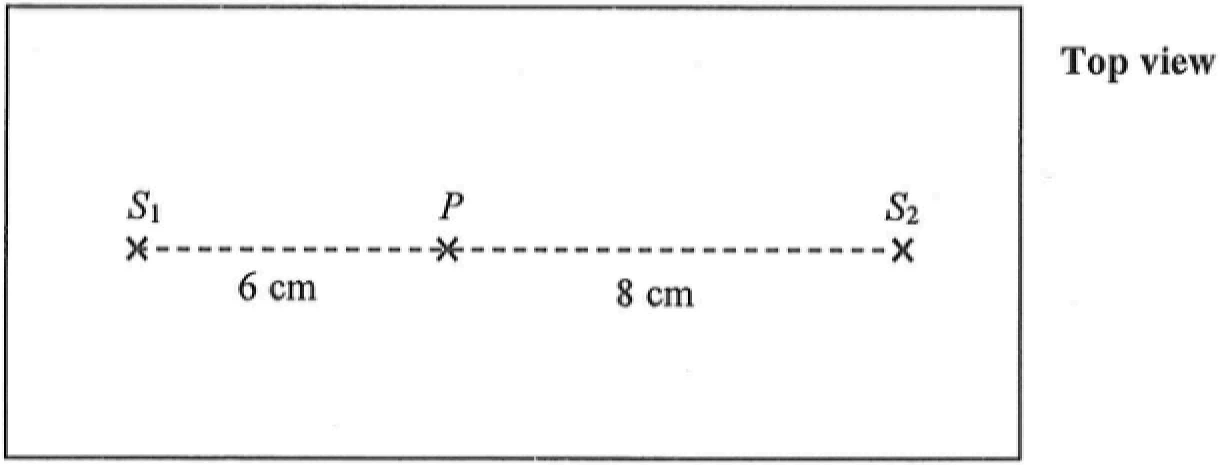


Two waves P and Q travel in the same direction and meet at a point. The above graph shows the variation of the displacement of each wave with time at that point. Which of the following statements is/are correct ?

(1) P and Q have the same frequency.
(2) The oscillation due to P is in anti-phase with that due to Q.
(3) The amplitude of the resultant wave at that point is four times the amplitude of P.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

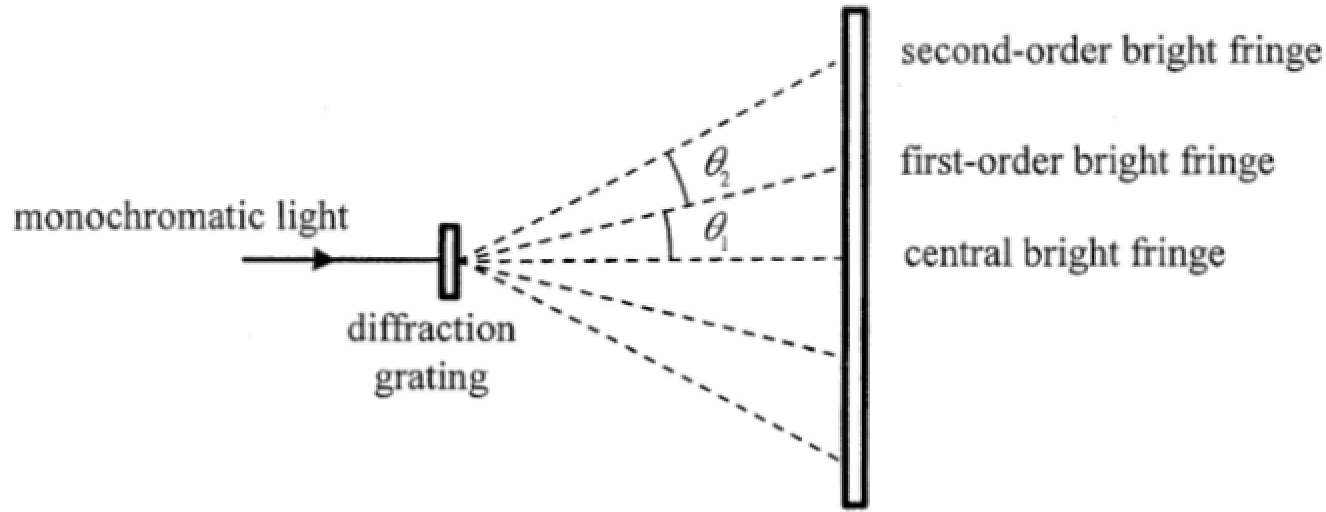
16. In the ripple tank shown below, dippers S_1 and S_2 are vibrating in phase and generate water waves of wavelength 0.8 cm towards each other at the same speed. For a particle P at distances 6 cm and 8 cm from S_1 and S_2 respectively as shown, the amplitude of the waves reaching P from S_1 is 1 mm while that from S_2 is 2 mm.



What is the amplitude of oscillation of P ?

A. 0 mm
B. 1 mm
C. 2 mm
D. 3 mm

*20. The figure below shows some of the bright fringes formed when monochromatic light passes through a diffraction grating.

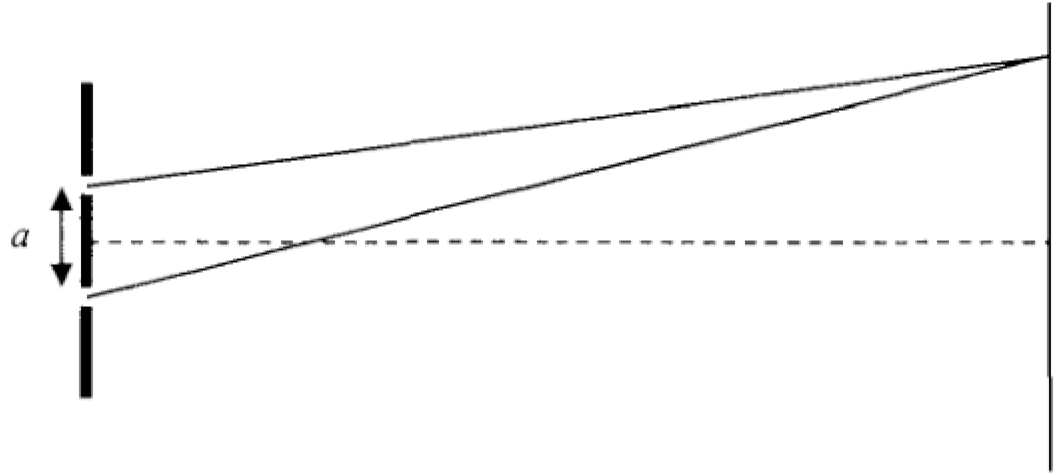


Which of the following is/are correct ?

(1) $\theta_1 = \theta_2$
(2) The maximum order of bright fringe is 4 if $\theta_1 = 20°$.
(3) θ_1 will decrease if the experiment is performed in water but not in air.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

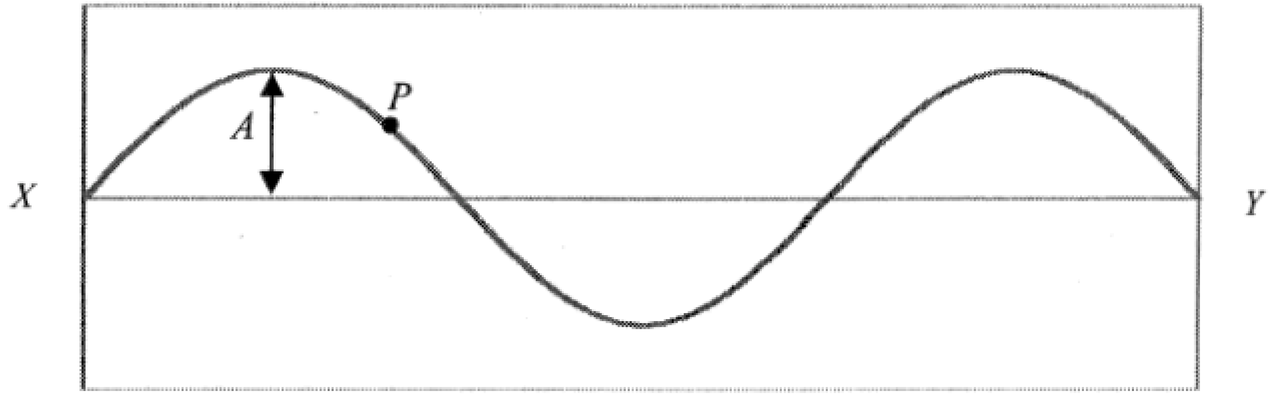
16. In Young's double slit experiment employing monochromatic light, how will the interference pattern change if the separation a of the double slit is reduced ?



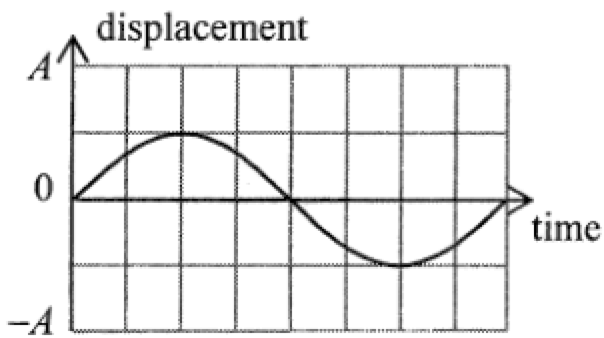
(1) The pattern will become brighter.
(2) The number of fringes that can be observed will increase.
(3) The fringe separation of the pattern will become larger.

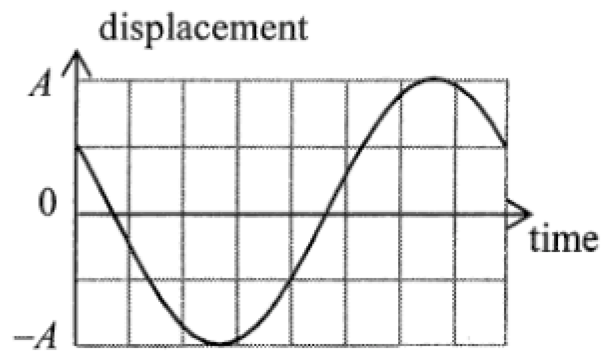
A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

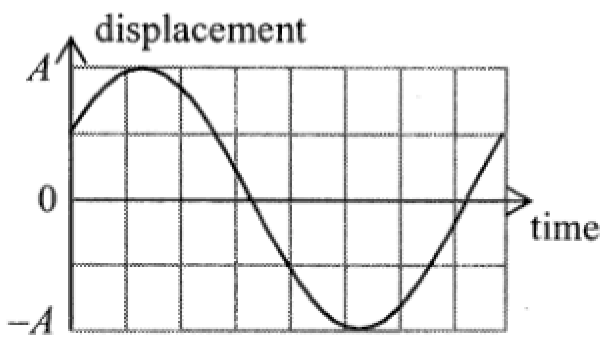
18. A stationary wave is formed on a string fixed at both ends X and Y. The following is a snapshot of the string at time $t = 0$. The amplitude of vibration at an antinode is A.

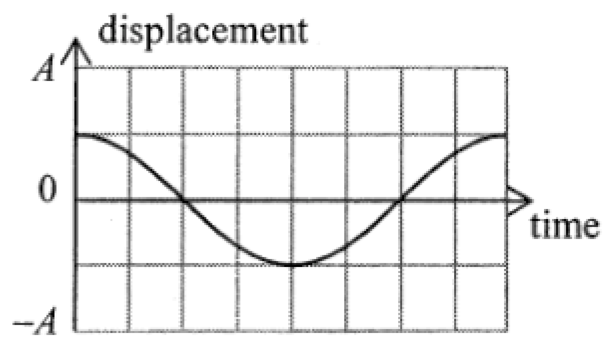


Which of the following shows the displacement-time graph of point P on the string for one period ? (Upward displacement is taken as positive.)

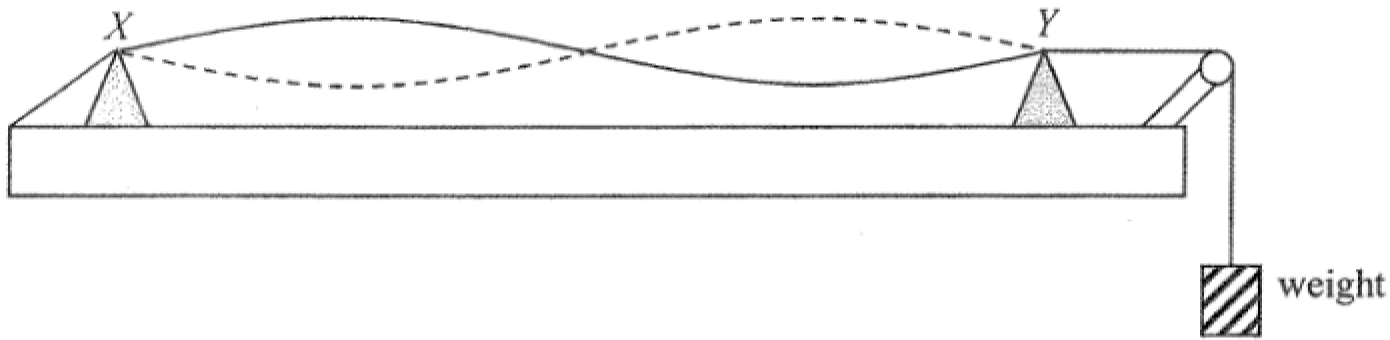
A.


B.


C.


D.


18.



A string is set to vibrate at frequency f such that a standing wave is formed between two fixed supports X and Y as shown. If the tension in the string is increased by adding weight gradually while the frequency is kept at f, which of the following is a possible mode of vibration at a steady state ?

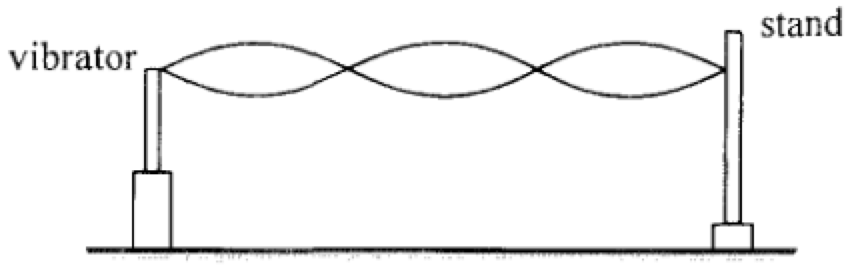
A. 

B. 

C. 

D. 

15. In the set-up below, different stationary wave patterns are formed on an elastic string by adjusting the frequency f of the vibrator.

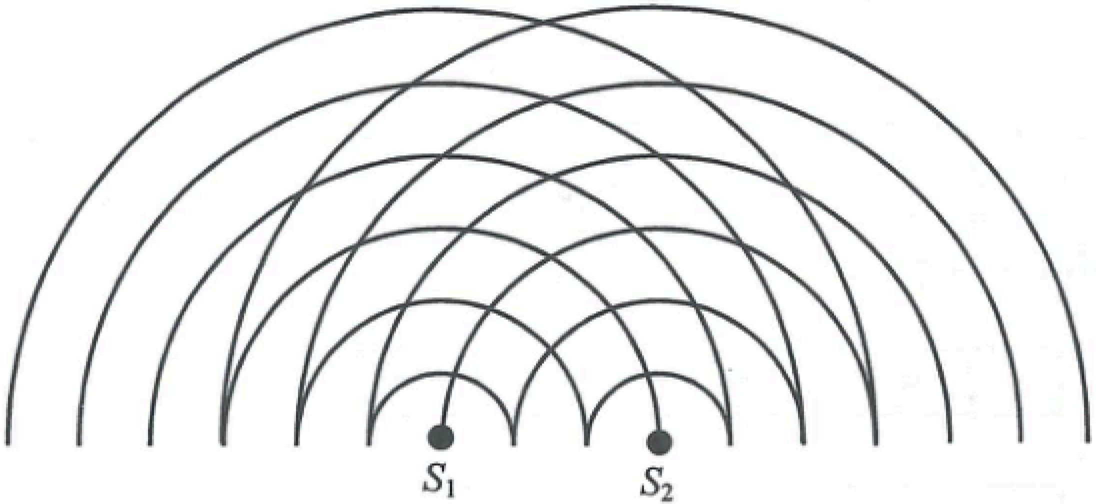


Which statements are correct when f increases ?

(1) The number of antinodes increases.
(2) The speed of the waves on the string increases.
(3) The frequency of the waves produced in air by the string increases.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

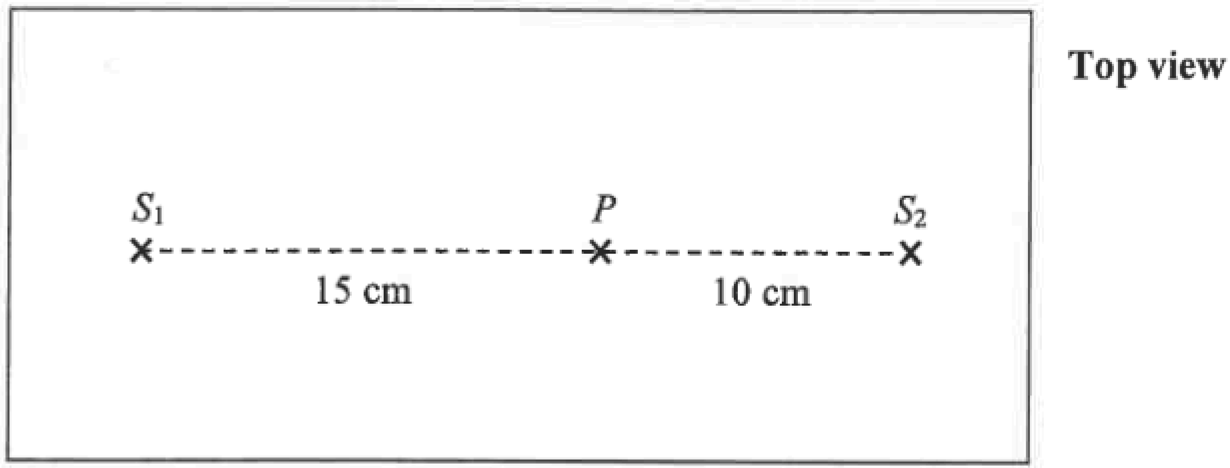
16.



The figure shows the circular water waves generated by two dippers S_1 and S_2 vibrating in phase. The lines represent wave crests. What is the number of nodal lines (i.e. minimum amplitude) formed ?

A. 3
B. 4
C. 6
D. 7

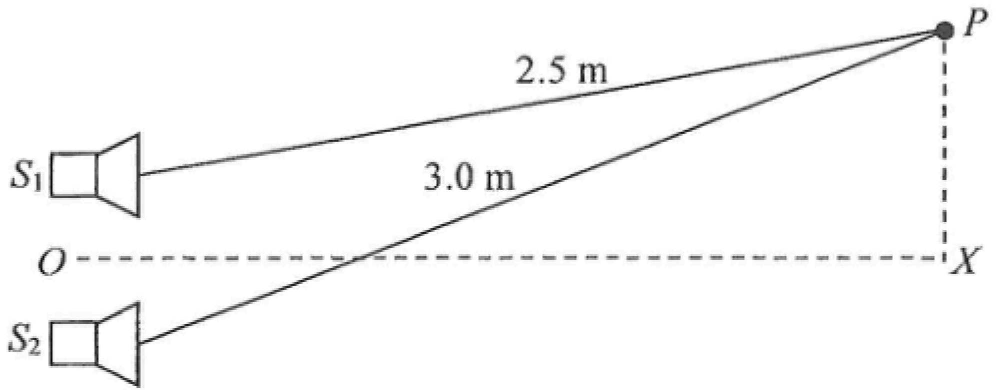
19. In the ripple tank shown below, dippers S_1 and S_2 are vibrating in phase with frequency f and generate two sets of waves towards each other at a speed of 20 cm s^{-1}. Destructive interference occurs at point P on the straight line S_1S_2.



Which of the following is a possible value of f ?

A. 24 Hz
B. 20 Hz
C. 18 Hz
D. 16 Hz

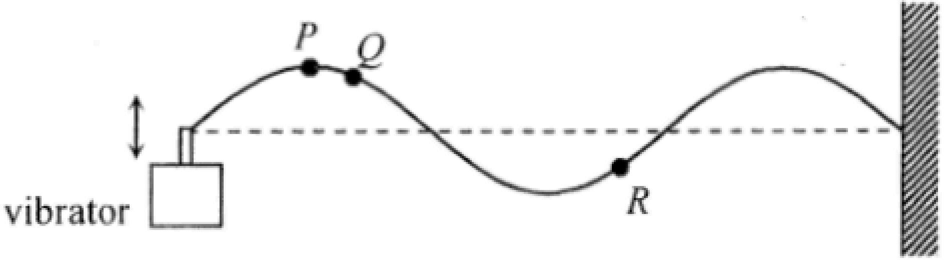
17. Two identical loudspeakers S_1 and S_2 connected to a signal generator produce sounds of wavelength 0.10 m which are in **anti-phase**.



OX is the perpendicular bisector of the line joining S_1 and S_2. What type of interference occurs at X and P respectively ?

	X	P
A.	destructive	constructive
B.	constructive	constructive
C.	destructive	destructive
D.	constructive	destructive

18.

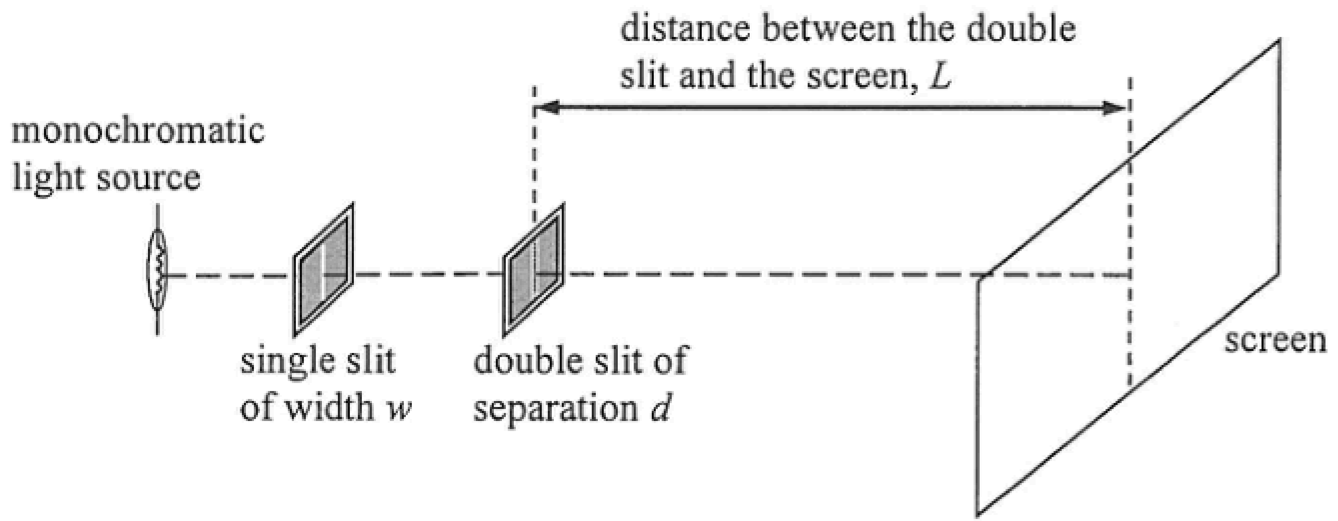


A vibrator generates a stationary wave on a string which is fixed at one end. The figure shows the appearance of the string at a certain instant. Which of the following descriptions about the motion of particles P, Q and R must be correct ?

(1) P and Q are momentarily at rest at this instant.
(2) Q and R take the same time to reach their respective equilibrium positions.
(3) P and R are always in antiphase.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

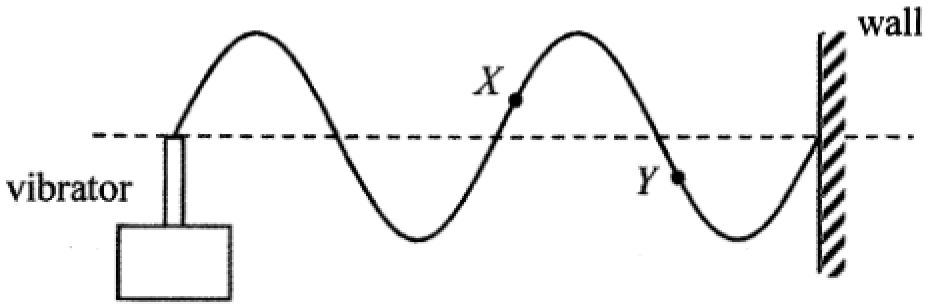
15. The figure shows a typical set-up of Young's double slit experiment.



Which combination below is the best setting for displaying an observable fringe pattern on the screen ?

	w	d	L
A.	0.1 mm	1 mm	10 m
B.	0.1 mm	1 mm	1 m
C.	1 mm	0.1 mm	1 m
D.	1 mm	0.1 mm	0.1 m

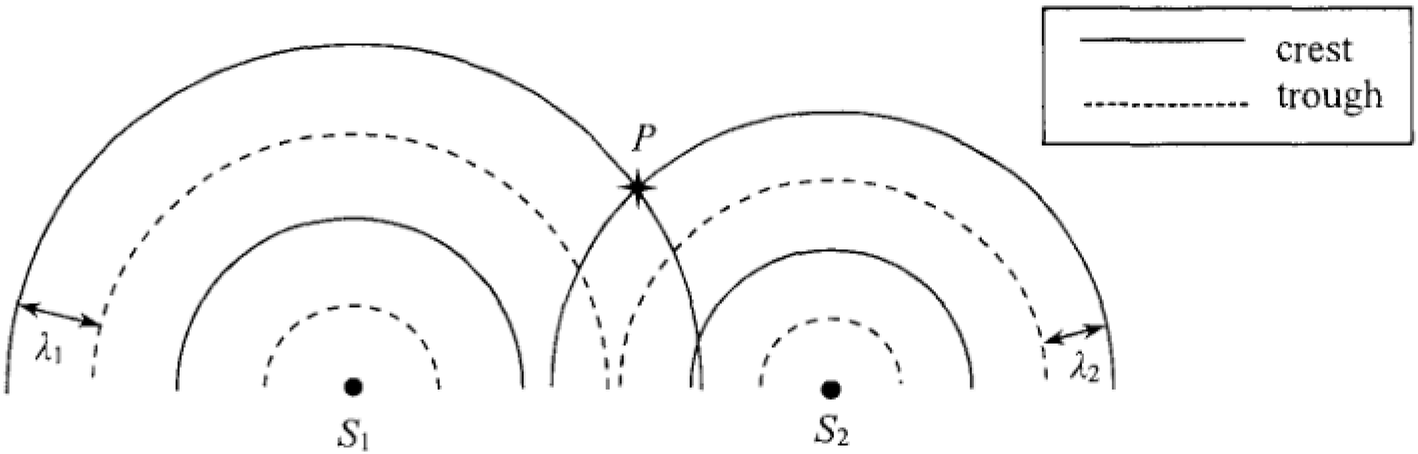
18. A string is tied to a vibrator while the other end is fixed to a wall. A stationary wave is formed as shown.



Which statement is correct when the frequency of the vibrator doubles ?

A. The wavelength will double.
B. The wave speed will double.
C. The amplitude will be halved.
D. Particles X and Y will become vibrating in phase.

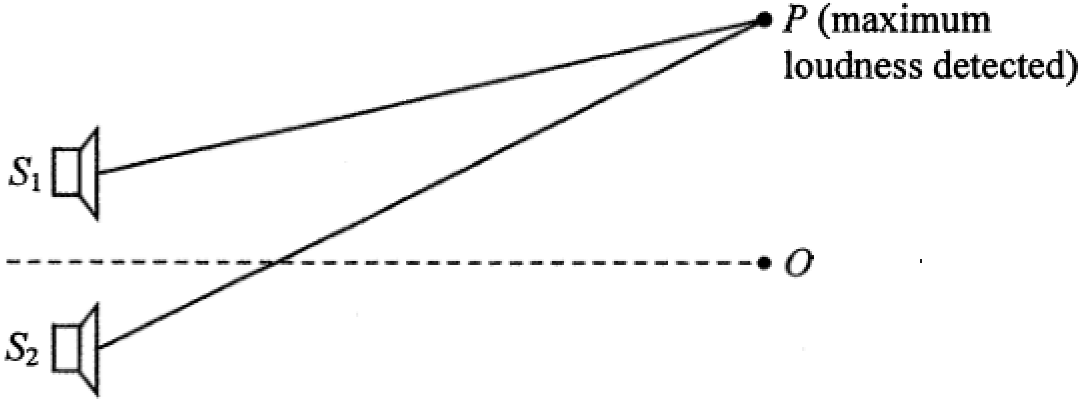
14. In a ripple tank, circular water waves of wavelengths λ_1 and λ_2 ($\lambda_1 > \lambda_2$) are produced by two vibrators S_1 and S_2 respectively. The figure represents the two circular waves propagating on the water surface at a certain moment.



Which of the following statements is correct ?

A. The particle at P is always at crest position.
B. At P, the two waves always reinforce to give a larger amplitude.
C. The principle of superposition cannot be applied at P as $\lambda_1 \neq \lambda_2$.
D. The principle of superposition can be applied at P but the two waves do not always reinforce at that location.

21.



Loudspeakers S_1 and S_2 connected to a signal generator emit sound waves which are in phase. Point O is equidistant from the loudspeakers while at point P maximum loudness is detected. The wavelength of the sound waves is λ. Which statement is **INCORRECT** ?

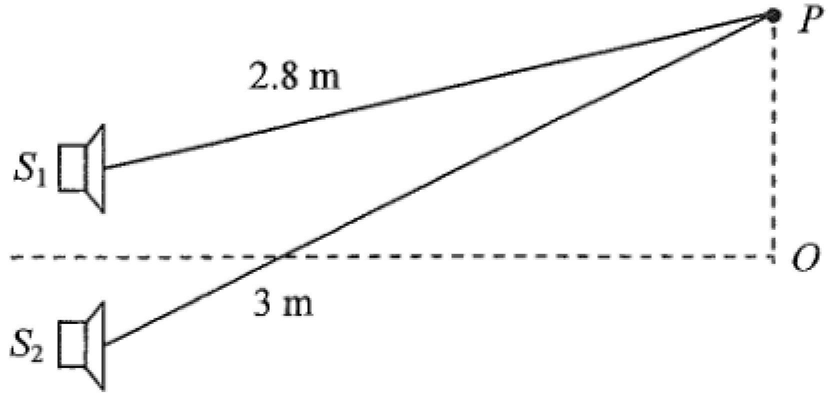
A. Both PS_1 and PS_2 must be integral multiples of wavelength λ.

B. The definite value of the path difference $PS_2 - PS_1$ cannot be determined from the information given.

C. At least one point of minimum loudness can be detected between O and P.

D. Minimum loudness will be detected at P if the sound waves from S_1 and S_2 are in antiphase.

18.



S_1 and S_2 are two loudspeakers connected to a signal generator but the sound waves produced by them are in anti-phase. Point O is equidistant from the loudspeakers while point P is at the distances shown in the figure from the loudspeakers. What type of interference occurs at O and P if the wavelength of the sound waves is 10 cm ?

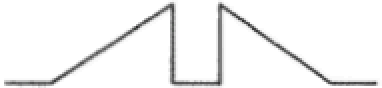
	O	***P***
A.	destructive	constructive
B.	constructive	constructive
C.	destructive	destructive
D.	constructive	destructive

14.

Two pulses of the same shape travel along a stretched string with one end fixed to the wall as shown above. Which of the following can be the resultant waveform at different instants later ?

(1)

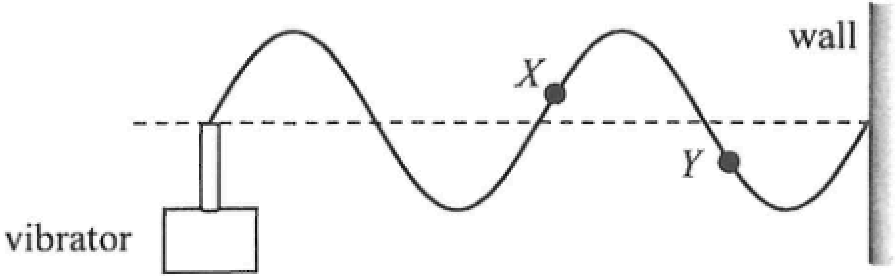
(2)



(3)

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

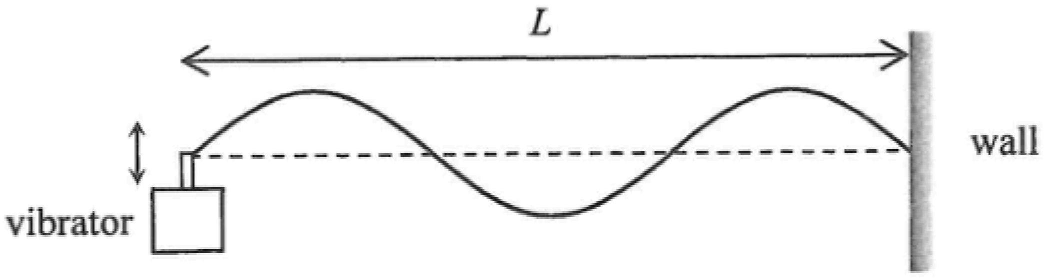
18. A string is tied to a vibrator at one end while the other end is fixed to a wall. The figure shows the stationary wave formed at a certain frequency of the vibrator.



Which statement is correct when the frequency is doubled ?

A. The amplitude of the wave will double.
B. The wavelength will double.
C. The speed of the wave will double.
D. Particle X and Y will become in phase.

13. The figure shows a string with one end fixed on the wall and the other end tied to a vibrator with frequency f. A stationary wave is formed as shown at a certain instant. The distance between the vibrator and the wall is L.



Which of the following statements are correct ?

(1) All particles on the string between two successive nodes are vibrating in phase.
(2) If the tension in the string increases while L remains unchanged, f must increase to maintain the same stationary wave pattern.
(3) The wavelength of the stationary wave is $\frac{L}{3}$.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

14. Which of the following phenomena involve(s) the principle of superposition of waves ?

(1) Ripples in a pond overlap and create regions with different amplitudes.
(2) Stationary wave patterns are formed when two waves of the same frequency travel in opposite directions.
(3) The speed of waves changes when they travel from one medium to another.

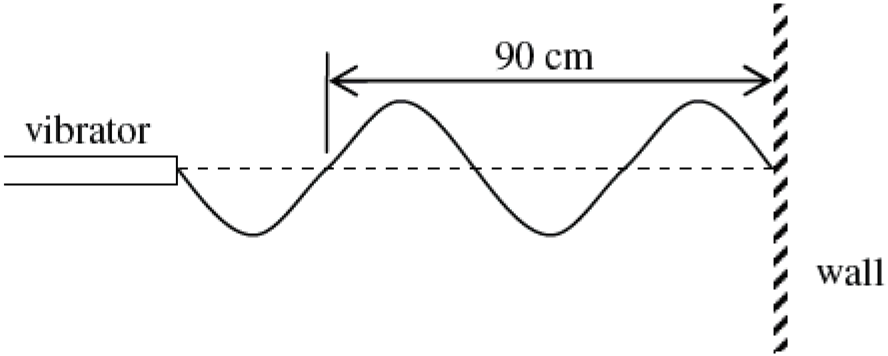
A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

19. A monochromatic light beam is directed to a double slit and a fringe pattern is formed. Which of the following physical phenomena is/are involved ?

(1) dispersion
(2) diffraction
(3) interference

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

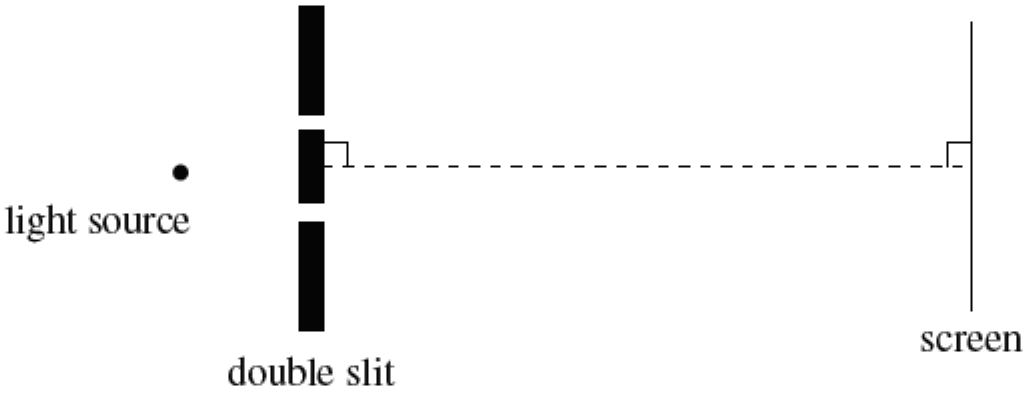
19.



A stationary wave is set up along a string by a vibrator. The waveform at a certain instant is shown above. If the frequency of the vibrator is 50 Hz, what is the wave speed along the string ?

A. 15 m s^{-1}
B. 30 m s^{-1}
C. 45 m s^{-1}
D. 55 m s^{-1}

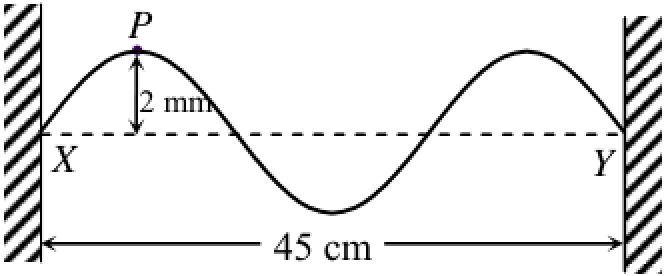
*22.



In a Young's double slit experiment, a monochromatic light source of wavelength 600 nm is used. The fringe separation is 5 mm on the screen. If the slit separation is halved and a monochromatic light source of wavelength 450 nm is used instead, what is the new fringe separation ?

A. 1.9 mm
B. 3.3 mm
C. 7.5 mm
D. 13.3 mm

19.



String *XY* is fixed at both ends. The distance between *X* and *Y* is 45 cm. Two identical sinusoidal waves travel along *XY* in opposite directions and form a stationary wave with an antinode at point *P*. The figure shows the string when *P* is 2 mm, its maximum displacement, from the equilibrium position. What is the amplitude and wavelength of each of the **travelling waves** on the string ?

	Amplitude	**Wavelength**
A.	1 mm	30 cm
B.	1 mm	15 cm
C.	2 mm	30 cm
D.	2 mm	15 cm

20. A Young's double-slit experiment was performed using a monochromatic light source. Which change would result in a greater fringe separation on the screen ?

(1) Using monochromatic light source of longer wavelength
(2) Using double slit with greater slit separation
(3) Using double slit with larger slit width

A. (1) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)